Accessible Quantum Natural Gradient: Optimization with Measurement-Accessible Tangent Geometry

Marwan Ait Haddou[1]

[1] *Cadi Ayyad University, Morocco*

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Natural-gradient methods precondition variational optimization with a state-space information metric, whereas practical quantum learning pipelines act through restricted measurements and low-dimensional classical readouts. We introduce the accessible quantum natural gradient (AQNG), which replaces the full quantum Fisher metric by the pullback of the score geometry visible through a specified measurement interface. For retained commuting readout features, the accessible metric is an exact projected-score Gram matrix, not merely a low-rank numerical approximation to the full QFIM. We evaluate AQNG in a frozen campaign of 470 paper-scale experiment cells, supplemented by 15 matched large/deep cells for Adam and SGD. Across 240 generic dataset–architecture cases, cross-fitted aligned AQNG has the lowest pooled mean test loss, but its advantage over random-rank AQNG and first-order baselines is modest and architecture dependent. In the matched large/deep follow-up, aligned AQNG outperforms SGD in all 15 cases and Full-QNG in 14 of 15, while its difference from Adam is not statistically resolved. Geometry and trainability also separate sharply: in SU(2)-Haar scaling, normalized aligned retention grows far above the random-rank baseline, yet optimization gains are nonmonotonic; in a number-conserving $U(1)$ control, accessibility remains near complete while the supervised gradient collapses by orders of magnitude; and increasing readout order can raise retained tangent mass while worsening optimization through severe metric conditioning. These results support accessible geometry as a controllable and diagnostically interpretable optimization resource, but not as a substitute for task-gradient signal or conditioning.

I. INTRODUCTION

The geometry of parameterized quantum states controls how changes in circuit parameters alter the represented state. For pure states, this local geometry is captured by the Fubini–Study metric, equivalently the quantum Fisher information matrix up to convention, and motivates quantum natural-gradient optimization [1, 2]. This viewpoint is attractive because it replaces Euclidean parameter distance by a state-sensitive notion of distance. It also raises a practical question that is usually left implicit: which part of the state-space geometry is actually visible to the learning interface used by the experiment?

A variational learner rarely observes the full state. It interacts with measurement outcomes, retained observables, marginals, or a small classical feature vector. Consequently, two tangent directions that are distinct in the full quantum metric can become indistinguishable after measurement and readout restriction. Conversely, a low-dimensional readout can be strongly aligned with a small set of task-relevant tangent directions even when it retains only a small fraction of the total state-space information. The distinction between full geometry and accessible geometry therefore matters both statistically and algorithmically.

The present work follows a sequence of increasingly operational geometric questions. First, Fubini–Study metric conditioning was used as a trainability control variable: the Sculpture framework meta-learns circuit initializations that improve the condition number of the full metric [3]. Second, under an isotropic tangent model, fixed-basis readout accessibility was shown to obey a rank law: the expected retained tangent mass is set by the readout rank relative to the score-space dimension [4]. Third, the isotropy assumption was removed by introducing a spectral orientation framework in which retained information depends jointly on the rank baseline, the tangent-covariance spectrum, and the relative orientation of the readout subspace [5]. The current paper closes the algorithmic loop: instead of using accessibility only as a diagnostic, we use the measurement-accessible tangent metric itself to precondition optimization.

There is related work on measurement-limited feature visibility from a different direction. Gross and Rieser formulate quantum extreme learning machines through Pauli-transfer matrices and characterize which encoded Pauli features are decodable through the row space selected by the reservoir dynamics and measurement operators [6]. Their problem is representational: a fixed quantum reservoir transforms an input feature library and a classical readout learns from the resulting features. Our problem is differential: the object being projected is the parameter-score tangent of a trainable circuit, and the projected geometry is subsequently used as an optimization metric. The common linear-algebraic skeleton is a projection onto an observable subspace, but the tangent object, normalization, learning dynamics, and operational question are different.

We introduce the *accessible quantum natural gradient* (AQNG). For a commuting retained readout, AQNG builds a covariance-normalized score Jacobian and uses its pullback Gram matrix as the preconditioner. This construction admits an exact projected-score interpretation: the accessible metric is the Gram matrix of the component of the full classical score that lies in the retained feature span. It is therefore not defined by approximating the full QFIM in matrix norm. Instead, the metric is tied to a physically specified measurement interface.

The empirical study is designed to separate three questions that are often conflated: whether a tangent direction is visible to the measurement, whether the resulting accessible metric is numerically well conditioned, and whether the supervised objective still carries a usable gradient signal. We use rank-matched physical, cross-fitted aligned, and random readouts; compare against full QNG, block QNG, Adam, and SGD; test analytic and finite-shot regimes; and include a number-conserving $U(1)$ control in which accessibility and trainability separate sharply. The central conclusion is deliberately narrower than a universal optimizer claim: accessible geometry can be a useful and interpretable optimization resource, but retained tangent information alone does not determine trainability.

The main contributions are:

1. an exact accessible-metric construction for retained commuting readouts, together with normalization, damping, and trust controls suitable for optimization;

2. a rank-matched experimental design that separates physical orientation, cross-fitted tangent alignment, and random-subspace baselines without changing readout rank;

3. a frozen multi-dataset benchmark and scaling study showing that AQNG can outperform full-QNG and SGD in specific regimes while remaining competitive rather than uniformly superior to Adam;

4. a diagnostic decomposition of optimization failure into measurement accessibility, metric conditioning, and task-gradient signal, illustrated by the readout-order and $U(1)$ controls.

II. MEASUREMENT-ACCESSIBLE TANGENT GEOMETRY

A. Score-space representation

Consider a parameterized state measured by a fixed commuting measurement with outcome probabilities $p_a(\boldsymbol{\theta})$ on a known support Ω. For outcome $a \in \Omega$, define the classical score

$$s_{ai} = \partial_i \log p_a, \tag{1}$$

and the probability-weighted score matrix

$$S_{ai} = \sqrt{p_a}\, s_{ai}. \tag{2}$$

The classical Fisher information matrix (CFIM) of the full measurement record is then

$$G_{\mathrm{C}} = S^{\mathsf{T}} S. \tag{3}$$

Because $\sum_a p_a s_{ai} = 0$, the score lives in the centered outcome subspace. For any fixed quantum measurement, $G_{\mathrm{C}} \preceq G_{\mathrm{Q}}$, where G_{Q} denotes the QFIM in the same convention.

Now retain r commuting readout functions $f_1, \ldots, f_r$ of the same outcomes. Let $F \in \mathbb{R}^{|\Omega| \times r}$ contain these functions and let

$$\widetilde{F}_{a\alpha} = F_{a\alpha} - \sum_b p_b F_{b\alpha} \tag{4}$$

be their centered version. Introduce

$$B = D_p^{1/2} \widetilde{F}, \qquad \Sigma = B^{\mathsf{T}} B, \tag{5}$$

where $D_p = \mathrm{diag}(p)$. The readout expectation Jacobian is

$$J = \partial_{\boldsymbol{\theta}} \langle F \rangle = B^{\mathsf{T}} S. \tag{6}$$

AQNG uses the covariance-normalized pullback

$$\boxed{G_{\mathrm{acc}} = J^{\mathsf{T}} \Sigma^{+} J.} \tag{7}$$

B. Exact projection identity

Equation (7) has a direct geometric interpretation. Substituting $J = B^{\mathsf{T}} S$ and $\Sigma = B^{\mathsf{T}} B$ gives

$$G_{\mathrm{acc}} = S^{\mathsf{T}} B (B^{\mathsf{T}} B)^{+} B^{\mathsf{T}} S \tag{8}$$

$$= S^{\mathsf{T}} P_B S, \tag{9}$$

where P_B is the orthogonal projector onto the probability-weighted centered readout span $\mathrm{col}(B)$. Thus, for commuting retained features, the accessible metric is exactly the Gram matrix of the projected measurement score. In particular,

$$0 \preceq G_{\mathrm{acc}} \preceq G_{\mathrm{C}} \preceq G_{\mathrm{Q}}. \tag{10}$$

The construction is therefore not a generic low-rank approximation to G_{Q}. It is the metric induced by a specified measurement interface.

The covariance normalization is essential. A change of coordinates $F \mapsto FM$ with nonsingular M leaves P_B, and hence G_{acc}, unchanged. The metric depends on the retained subspace rather than on arbitrary rescalings of the readout basis.

C. Retention and orientation

To diagnose which score directions a rank-r readout captures, we use a normalized tangent-score covariance $C \succeq 0$ with $\mathrm{Tr}\, C = 1$. If P projects onto the retained score subspace, the retained mass is

$$R = \mathrm{Tr}(PC), \tag{11}$$

and the rank-normalized retention is

$$\rho = \frac{R}{r/N}, \tag{12}$$

where N is the centered score-space dimension. Under a uniformly random rank-r subspace, $\mathbb{E}\rho = 1$ [4, 5]. Values $\rho \gg 1$ therefore diagnose orientation beyond the rank baseline. The spectral theory further shows that random-subspace fluctuations are controlled by the spectrum of C, so rank, anisotropy, and orientation must be separated rather than collapsed into a single notion of "information retained" [5].

D. Scope

The exact projector identity above assumes retained functions that are jointly estimable from one commuting measurement record. Noncommuting observables require an explicit grouping or measurement model; a covariance matrix written for incompatible observables does not by itself define one operational interface. All primary AQNG experiments in this paper use diagonal functions of computational-basis outcomes, so the commuting construction applies directly.

III. ACCESSIBLE QUANTUM NATURAL GRADIENT

Given a supervised loss $L(\boldsymbol{\theta})$ with Euclidean gradient $g = \nabla_{\boldsymbol{\theta}} L$, AQNG computes a damped preconditioned direction

$$d_{\mathrm{A}} = (\widehat{G}_{\mathrm{acc}} + \lambda I)^{-1} g, \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta d_{\mathrm{A}}. \tag{13}$$

The full-QNG control replaces $\widehat{G}_{\mathrm{acc}}$ by the full QFIM. In the implementation, the accessible metric can be factorized as $A^{\mathsf{T}}A$, allowing primal, dual, or SVD-based solves depending on parameter and readout dimensions. Metric evaluations are cached for a fixed number of optimization steps to separate metric-construction cost from the loss-gradient loop.

A. Metric normalization and damping

Comparisons between metrics with different overall scales are otherwise confounded by damping. The primary configuration therefore normalizes each metric by its trace before applying a common absolute damping coefficient. Additional controls use no normalization or maximum-eigenvalue normalization and alternative damping conventions based on the mean or maximum eigenvalue. We also impose a Euclidean direction cap and an accessible-metric trust radius. These controls prevent a nearly singular metric from converting a small gradient into an arbitrarily large parameter step. Consequently, update magnitude is not used as evidence that a vanishing-gradient problem has been solved.

B. Rank-matched readout designs

The primary readout comparison keeps rank fixed and changes only orientation. The physical readout is the local computational-basis Z span. A random control draws a subspace of exactly the same rank from the centered outcome-function space. The aligned control is fit from an independent calibration set of tangent scores: the leading rank-matched score directions are estimated on calibration tangents and then frozen before optimization and evaluation. Labels are not used in this calibration step. This cross-fit construction avoids evaluating an aligned subspace on the same tangent sample used to fit it.

The rank-matched comparison is important because increasing the number of retained observables changes both dimension and orientation. By holding rank fixed, physical, random, and aligned AQNG test whether orientation relative to the tangent covariance changes optimization at a fixed readout dimension. A separate readout-order ablation deliberately changes rank by moving from weight-one diagonal Pauli functions to all diagonal strings through weight two.

C. Finite-shot estimation

In finite-shot mode, all retained diagonal features are derived from the same computational-basis bitstrings. Known outcome support is fixed structurally rather than inferred from observed nonzero counts. For the number-conserving control, the support is the corresponding fixed-Hamming-weight sector. A symmetric Dirichlet pseudocount regularizes empirical probabilities on that known support before constructing score and covariance estimates. This prevents unobserved-but-allowed outcomes from producing artificial support collapse at finite shot count.

The finite-shot Full-QNG control uses an analytic full metric while its loss is sampled. It is therefore an optimizer reference, not a hardware-resource-matched baseline. Shot accounting for AQNG reports calibration and training resources separately.

IV. EXPERIMENTAL PROTOCOL

A. Frozen benchmark matrix

The paper-scale campaign contains 470 completed experiment cells and 3160 terminal optimizer rows. The primary benchmark is the Cartesian product of four binary datasets (Iris, Breast Cancer, Wine, and binary

Digits), four six-qubit circuit families (RyRz–CZ, SU(2)–CNOT, SU(2)–Haar, and a number-conserving $U(1)$ family), and 20 paired stochastic seeds, giving 320 dataset–architecture cells. Each primary cell contains eight optimizers: physical-, random-, and aligned-readout AQNG; an AQNG-Z_0 minimal-readout control; full QNG; block QNG; SGD; and Adam.

The auxiliary matrix contains 40 qubit-scaling cells, 30 depth-scaling cells, 40 readout-order cells, and 40 finite-shot cells. Qubit scaling uses SU(2)–Haar and $U(1)$ circuits at $n \in \{4, 6, 8, 10\}$ with five seeds. Depth scaling uses the same two families at depths $L \in \{1, 3, 5\}$ with five seeds. The readout-order ablation uses Iris and Digits with SU(2)–Haar and $U(1)$, comparing weight-one diagonal readouts with the complete diagonal Pauli span through weight two. Finite-shot experiments use the same two datasets and two circuit families at 1000 and 10000 shots.

After the full matrix had been inspected, a separate follow-up added the missing Euclidean baselines for the larger/deeper SU(2)–Haar settings: Adam and SGD at $(n, L) = (8, 3), (10, 3), (6, 5)$ with the same five paired seeds. This adds 15 experiment cells and 30 terminal baseline rows. Because this extension was specified after the main outcomes were known, we report it explicitly as a follow-up rather than merging it silently into the original frozen design.

B. Shared training controls

Within a paired comparison, methods share the same dataset split, circuit family, initialization seed, training budget, and loss definition. The paper-scale runs use 20 optimization steps, a loss minibatch of four samples, a metric minibatch of two samples, and a metric refresh every two steps. The default rank-matched orientation experiment uses a weight-one diagonal readout. Cross-fitted aligned readouts are calibrated on independent tangent samples and then frozen before supervised training.

AQNG orientation methods use learning rate 0.06 and damping 0.003 under trace metric normalization and absolute damping. Baseline learning rates were selected in a restricted tuning stage using only Iris/SU(2)–Haar seed 0: 0.10 for SGD and 0.06 for Adam; AQNG-Z_0 and block-QNG use the same 0.06/0.003 pair. The tuning stage excludes the $U(1)$ control and does not inspect test outcomes from the paper-scale matrix.

C. Statistical reporting

The unit of pairing is the stochastic seed within a fixed dataset–architecture–protocol cell. Primary method comparisons use paired Wilcoxon signed-rank tests and Holm correction over the planned family of comparisons. We report pooled means only as descriptive aggregates unless a corresponding paired test is defined. The large/deep follow-up reports both setting-wise five-seed comparisons and a pooled 15-case paired analysis; the latter is useful as a compact summary but does not replace the setting-wise uncertainty.

All paper-facing tables are regenerated from the committed aggregate files in `results/paper/paper_scale_v2/`. The source paper-scale workflow contains 470 cells, 3160 terminal rows, and 470 calibration rows; the follow-up contains 15 cells and 30 terminal rows. No result in the manuscript is reconstructed from the superseded AQNG draft.

TABLE I. Pooled generic primary benchmark over 240 cases. Lower test loss and higher accuracy are better.

Method	Test loss	Accuracy
AQNG-aligned	**0.5945**	**0.7947**
AQNG-random	0.5956	0.7914
SGD	0.6084	0.7858
Adam	0.6203	0.7731
AQNG-physical	0.6229	0.7878
AQNG-Z_0	0.6352	0.7828
Full-QNG	0.6562	0.7689
Block-QNG	0.6781	0.7606

V. RESULTS

A. Generic primary benchmark

We first pool the three nonconserving circuit families across the four datasets and 20 seeds, giving 240 paired cases per method. Table I reports descriptive pooled means. Cross-fitted aligned AQNG has the lowest mean test loss and the highest mean test accuracy, but the margin over random-rank AQNG is small. The result therefore supports useful accessible preconditioning without supporting a universal claim that tangent alignment is always the best readout design.

The pooled ordering should not be read as a total optimizer ranking. Cellwise outcomes remain architecture and dataset dependent, and the gap between aligned AQNG, random AQNG, SGD, and Adam is much smaller than the gap to full- and block-QNG in this frozen configuration. The main conclusion from the primary matrix is therefore comparative rather than absolute: an accessible metric can preserve competitive optimization performance while exposing which measurement subspace supplies the preconditioning geometry.

B. Rank baseline and qubit scaling

The scaling experiment gives a clean geometric check of the readout-rank theory. For SU(2)–Haar circuits, the rank baseline r/N decreases rapidly with n, while a random rank-matched readout stays near $\rho = 1$. In contrast,

TABLE II. SU(2)–Haar scaling geometry.

n	r/N	R_{aligned}	ρ_{aligned}	ρ_{random}
4	0.2667	0.4319	1.62	1.00
6	0.0952	0.3606	3.79	1.04
8	0.0314	0.3460	11.03	1.04
10	0.0098	0.2997	30.66	0.99

TABLE III. Mean test loss in the matched large/deep follow-up.

Method	$n = 8, L = 3$	$n = 10, L = 3$	$n = 6, L = 5$
AQNG-aligned	**0.168**	0.254	0.139
AQNG-random	0.210	0.252	**0.127**
AQNG-physical	0.238	**0.241**	0.183
Adam	0.231	0.265	0.184
SGD	0.425	0.407	0.367
Full-QNG	0.476	0.430	0.388

both the physical and cross-fitted aligned readouts become increasingly super-baseline. The aligned normalized retention grows from approximately 1.62 at $n = 4$ to 30.66 at $n = 10$.

This strong geometric orientation does not translate into a monotonic optimization advantage. The aligned mean test losses at $n = 4, 6, 8, 10$ are 0.221, 0.237, 0.168, and 0.254, respectively. The corresponding physical-readout losses are 0.207, 0.273, 0.238, and 0.241. Thus aligned AQNG is best at $n = 6$ and $n = 8$, but not at $n = 4$ or $n = 10$. Accessibility can therefore increase relative to the null baseline while task utility remains nonmonotonic.

C. Matched large/deep comparison with Adam and SGD

The post-hoc follow-up fills the missing first-order baselines at $(n, L) = (8, 3), (10, 3), (6, 5)$ for SU(2)–Haar on Iris. Table III combines the matched five-seed means. The pooled 15-case loss is 0.1867 for aligned AQNG, compared with 0.1961 for random AQNG, 0.2207 for physical AQNG, 0.2268 for Adam, 0.3998 for SGD, and 0.4312 for full QNG.

In the pooled paired analysis, aligned AQNG beats SGD in 15/15 cases and full QNG in 14/15. The Holm-adjusted Wilcoxon values are $p_{\text{Holm}} = 1.22 \times 10^{-3}$ against SGD and 5.80×10^{-3} against full QNG. Against Adam, aligned AQNG wins 10/15 cases with a mean loss advantage of 0.0400, but the Holm-adjusted test is not significant. The correct interpretation is therefore that AQNG is clearly stronger than SGD and full QNG in this matched follow-up, while Adam remains a competitive baseline.

D. More retained geometry can hurt

The readout-order ablation provides a direct counterexample to the idea that maximizing retained tangent mass is sufficient. On Iris/SU(2)–Haar, increasing the diagonal readout from weight one to weight two raises aligned retention from 0.361 to 0.797, yet mean test loss worsens from 0.237 to 0.361. On Digits/SU(2)–Haar, aligned retention rises from 0.308 to 0.738, while loss worsens from 0.620 to 0.716. The first-step quantity $g^{\mathsf{T}} d$ simultaneously drops from 7.05 to 2.57 on Iris and from 3.82 to 1.71 on Digits. Metric diagnostics show that the richer readout is substantially harder to condition. These results motivate condition-aware rather than retention-maximizing readout design.

E. Accessibility does not imply trainability

The number-conserving $U(1)$ family is the strongest negative control. Its physical and aligned readouts remain nearly maximally accessible as system size grows: at $n = 8$, aligned retention is 0.99993, and at $n = 10$ it is 0.99987. Nevertheless, all tested optimizers approach the same chance-like accuracy of 0.4667 at $n = 8$ and $n = 10$, with terminal loss near 2.1333. At the same time, the recorded first-step descent signal collapses by many orders of magnitude. The failure is therefore not measurement blindness. The task gradient itself becomes too weak to supply useful supervised information.

This regime is consistent with a barren-plateau-like gradient-collapse signal, but we do not claim a formal barren-plateau scaling theorem: the scaling study uses five seeds and was not designed to estimate an exponential gradient-variance law over a large initialization ensemble. The result needed here is narrower and directly observed: near-complete measurement accessibility does not rescue a vanishing task-gradient regime.

F. Finite-shot robustness

At 1000 and 10000 shots, AQNG remains operational across the tested Iris and Digits SU(2)–Haar cells. For example, on Digits the aligned losses are 0.661 and 0.624 at 1000 and 10000 shots, respectively, compared with 0.829 and 0.790 for the full-QNG control. On Iris, random-rank AQNG is the best of the tested AQNG variants at both shot counts. The ordering is therefore not monotonic in shot count or readout orientation. Because the finite-shot full-QNG implementation uses an analytic metric oracle, these numbers are optimization references rather than a resource-fair hardware comparison.

VI. GEOMETRIC INTERPRETABILITY

AQNG exposes diagnostics that are hidden by a purely Euclidean optimizer. The relevant notion of interpretability is geometric and mechanistic rather than semantic: the metric identifies which parameter-score directions are visible to the retained measurement interface, how strongly those directions are weighted, and whether the resulting preconditioner is numerically usable.

Three quantities separate distinct failure modes. First, retention R and normalized retention ρ diagnose *accessibility*: whether the readout sees the tangent covariance beyond its rank baseline. Second, the spectrum of G_{acc} diagnoses *conditioning*: whether the visible geometry can be inverted stably after normalization and damping. Third, the raw task-gradient norm or finite-shot signal-to-noise ratio diagnoses *supervised signal*: whether the objective supplies enough directional information for any preconditioner to exploit.

The experiments instantiate all three regimes. SU(2)–Haar scaling produces strongly super-baseline accessible subspaces and competitive optimization. The weight-two readout ablation has substantially higher retention but worse optimization, revealing a conditioning bottleneck. The $U(1)$ scaling control has nearly complete accessibility but a collapsing task signal, revealing a trainability bottleneck that no accessible metric can repair by itself. In this sense, AQNG is not only a preconditioner; it provides a decomposition of why optimization succeeds or fails.

This diagnostic interpretation should not be confused with semantic feature attribution. AQNG does not by itself imply that an eigenvector of the accessible metric corresponds to a human-interpretable concept or to one classical input feature. A semantic layer becomes possible only when the retained observables or tangent directions are tied to a structured feature dictionary with external meaning. The Pauli-transfer framework of Gross and Rieser provides an example of such a representational interpretation for encoded Pauli features in QELMs [6]. Extending AQNG in that direction would require an explicit map from accessible tangent directions to known data features or physically meaningful observables; this is outside the claims of the present study.

VII. DISCUSSION AND CONCLUSION

The experiments support a specific claim: measurement-accessible tangent geometry can be used as a physically grounded preconditioner, and in several regimes it is more useful than the full state-space metric or a Euclidean gradient. They do not support the stronger claim that AQNG is a universally superior optimizer. Random-rank AQNG is often close to aligned AQNG, Adam remains competitive in the large/deep follow-up, and the best readout orientation depends on architecture, depth, and system size.

The distinction between *accessibility* and *trainability* is central. A readout can capture nearly all of the relevant tangent covariance while the supervised gradient is effectively absent, as in the large-n $U(1)$ control. Conversely, a readout can retain more tangent mass while producing a less useful preconditioner because the metric becomes poorly conditioned, as in the weight-two ablation. Retention is therefore a resource descriptor, not an optimization objective by itself. A more complete trainability picture requires at least three ingredients: accessible orientation, stable metric spectrum, and surviving task-gradient signal.

This also clarifies the relation to barren plateaus. AQNG rescales and rotates an observed gradient; it cannot create supervised information that has already fallen below the estimation noise floor. In an exact arithmetic model, an inverse metric can compensate some common scaling between gradient and geometry, but finite damping, small-eigenvalue instability, and metric-estimation noise limit that mechanism. Demonstrating barren-plateau mitigation would require a dedicated scaling study of gradient variance or update signal-to-noise with substantially larger initialization ensembles and unclipped update diagnostics. The present $U(1)$ experiment is therefore reported as a gradient-collapse control rather than as a formal barren-plateau theorem.

Several resource caveats are also important. Cross-fitted alignment requires calibration tangents and those measurements are not free; the repository records calibration separately from training. The finite-shot full-QNG control uses an analytic metric oracle, so it is not a hardware-cost-matched comparator. The primary 20-seed benchmark keeps the dataset split fixed, meaning that it probes initialization and optimization stochasticity rather than split-to-split generalization uncertainty. Finally, the exact projector identity used by AQNG assumes a jointly measurable commuting readout; extending the method to noncommuting observable families requires an explicit measurement-grouping model.

The readout-order failure suggests a concrete next algorithmic step. Rather than choosing the subspace that maximizes retained tangent mass, one can optimize a regularized criterion that trades accessibility against spectral conditioning, for example through eigenvalue truncation, adaptive damping, or a condition-number penalty. A second direction is semantic structure: if the retained observables are tied to known input features or physical quantities, accessible eigendirections may be annotated with domain meaning rather than only geometric meaning. A third is a dedicated trainability-scaling experiment in which the raw gradient, accessible spectrum, metric-estimation noise, and shot complexity are tracked simultaneously.

In summary, AQNG converts measurement-accessible tangent geometry from a diagnostic into an optimizer. Its main value is not that it always beats first-order methods, but that it makes the geometry used for preconditioning operationally tied to the measurement interface and diagnostically separable into visibility, conditioning, and task signal. Across the frozen benchmark, that geometry is often sufficient for competitive or improved optimization, while the counterexamples show exactly where accessibility alone stops being predictive. The resulting principle is that useful quantum optimization geometry is not determined by retained information alone, but jointly by orientation, conditioning, and surviving gradient signal.

Appendix A: Implementation details

The reference implementation uses PennyLane 0.45.x with Autograd-compatible circuit derivatives. The accessible metric is formed from retained computational-basis outcome functions and supports primal, dual, and singular-value-based linear solves. In the paper-scale configuration, metrics are trace-normalized, damped with an absolute coefficient $\lambda = 0.003$, refreshed every two optimization steps, and combined with a Euclidean direction cap and an accessible-metric trust radius. These controls are fixed before the paper-scale outcomes are inspected.

The primary physical readout uses local Z functions. Rank-matched aligned and random controls have exactly the same retained dimension. The aligned basis is estimated from 64 calibration tangents and evaluated on an independent set of 64 held-out tangents before being frozen for training. The order-two ablation replaces the weight-one span by all diagonal Pauli strings through weight two.

For finite-shot probability estimates, the support is fixed from circuit structure. Generic circuits use the full computational-basis support. Number-conserving circuits use the known fixed-Hamming-weight sector. A symmetric Dirichlet pseudocount of $1/2$ is added on the known support before score construction, and the readout covariance receives an additional numerical regularization term. This choice prevents empirical zero counts from changing the model support.

All durable experiment outputs, aggregate tables, configuration records, and checksums are committed under `results/paper/paper_scale_v2/`. The report records the source workflow revisions for both the 470-cell paper-scale campaign and the 15-cell SGD/Adam follow-up.

Appendix B: Statistical reporting

Primary hypothesis tests use paired seeds within a fixed dataset–architecture cell. For each planned method comparison, the statistic is computed from the paired terminal test-loss differences using the Wilcoxon signed-rank test. The resulting family of p-values is corrected with the Holm procedure. Means, standard deviations, win counts, and pooled summaries are reported alongside corrected tests because the optimization distributions can be heterogeneous across circuit families.

The large/deep Adam/SGD extension is explicitly post-hoc relative to the completed paper-scale campaign. It uses the previously frozen learning rates and the same paired seeds at the three selected SU(2)–Haar settings. We report both the five-seed setting-wise comparisons and a pooled 15-case paired summary. The pooled test is treated as a follow-up aggregate rather than as a preregistered primary endpoint.

The qubit-scaling $U(1)$ study is not used to estimate an asymptotic barren-plateau exponent. Its five-seed design is sufficient to demonstrate the observed separation between near-complete readout accessibility and a collapsing supervised optimization signal, but not to establish an exponential gradient-variance law. Any future barren-plateau claim should be based on a dedicated random-initialization ensemble, explicit scaling fits, and finite-shot signal-to-noise or shot-complexity measurements.

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